1. (a) $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$
   (b) $P(A^c \cap B) = P(B) - P(A \cap B) = 0.4 - 0.2 = 0.2$
   (c) $P((A \cup B)^c) = 1 - 0.7 = 0.3$

2.
$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - (0.3 + 0.6 - 0.1) = 0.2$$

3. (a) $P(A \mid B) = \frac{0.35}{0.5} = 0.7$
   (b) $P(B \mid A) = \frac{0.35}{0.7} = 0.5$

4. Check independence:
$$P(A)P(B) = 0.4 \cdot 0.5 = 0.2 \neq 0.25$$
   Not independent.

5. There are 12 face cards, 3 of which are hearts.
$$P(A \mid B) = \frac{3}{12} = \frac{1}{4}$$

6.
$$\frac{\binom{13}{2}\binom{39}{3}}{\binom{52}{5}}$$

7. Given exactly 2 hearts:
$$\frac{\binom{3}{2}}{\binom{13}{2}} = \frac{3}{78} = \frac{1}{26}$$

8.
$$\sum_{x=0}^{3} c(x^2 + x) = c(0 + 2 + 6 + 12) = 20c = 1 \Rightarrow c = \frac{1}{20}$$
$$P(X \geq 2) = p(2) + p(3) = \frac{6}{20} + \frac{12}{20} = \frac{18}{20} = \frac{9}{10}$$

9.
$$E[X] = \sum xp(x) = \frac{1}{20}(0 + 2 + 12 + 36) = \frac{50}{20} = 2.5$$

10.
$$E[X^2] = \frac{1}{20}(0 + 2 + 24 + 108) = \frac{134}{20} = 6.7$$
$$\mathrm{Var}(X) = 6.7 - (2.5)^2 = 0.45$$

11. (a) $P(Y \leq 2) = 0.1 + 0.2 = 0.3$

    (b) $E[Y] = 0.1(1) + 0.2(2) + 0.3(3) + 0.4(4) = 3$

    (c) $E[Y^2] = 0.1 + 0.8 + 2.7 + 6.4 = 10$

$$\text{Var}(Y) = 10 - 9 = 1$$

12. (a) Example: Flip a fair coin twice. A = first flip is heads, B = second flip is heads.

$$P(A) = P(B) = \frac{1}{2}, \quad P(A \cap B) = \frac{1}{4}$$

    Independent, not mutually exclusive.

    (b) Example: Single die roll. $A = \{1\}$, $B = \{2\}$

$$P(A) = P(B) = \frac{1}{6}, \quad P(A \cap B) = 0$$

    Mutually exclusive, not independent.